

Barry Cox momentum project!!

Introduction

The purpose of this project was to examine whether a momentum trading approach can produce returns. Momentum trading is based on the idea that stocks that have recently performed well may continue to do well, while stocks that have performed poorly may continue to fall. This strategy was tested and compared against the QQQ exchange-traded fund.

Choice of ETF The ETF selected for this analysis was QQQ. QQQ follows the Nasdaq-100 Index and includes more than 100 large companies, mainly in technology and growth sectors. The 100 stocks is enough for this project.

The Data Collection Process

Price data was gathered for all stocks included in the QQQ ETF. The dataset covered roughly five years. Daily prices were converted into monthly prices by using the closing price from the final trading day of each month.

Momentum Measurement

Momentum was measured using three time periods: three months, six months, and twelve months. Each of these returns shows how much a stock's price changed over time. The three momentum values were averaged to form a single momentum score used for ranking stocks.

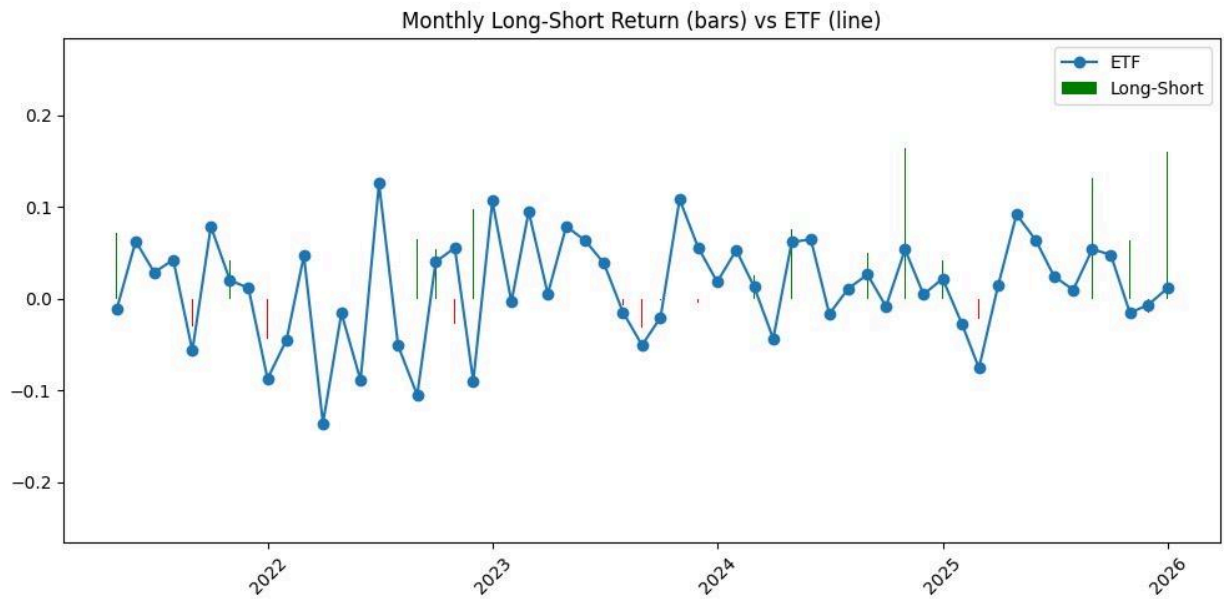
My portfolio setup

At the end of each month, stocks were ranked based on their momentum score. The 15 highest-ranked stocks were selected for the long portfolio, while the 15 lowest-ranked stocks were placed in the short portfolio. The portfolio was adjusted every month.

A historical backtest was conducted to evaluate how the momentum strategy would have performed in the past. The strategy was updated monthly, and returns were calculated for the long side, short side, and the combined long-short portfolio. These results were compared to QQQ returns.

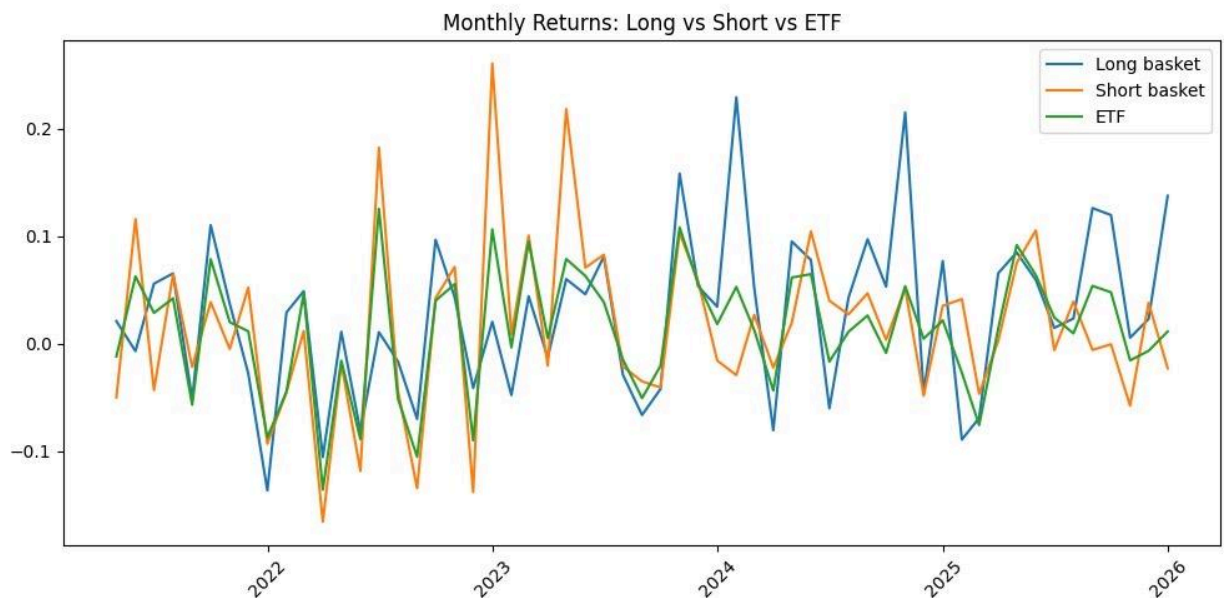
Explanation of Charts

Chart 1: Monthly Long-Short Performance Compared to ETF



This chart displays the monthly gains and losses of the long-short strategy using bars, while a line represents the ETF's monthly return. It allows for an easy comparison between the strategy and the benchmark each month.

Chart 2: Monthly Returns of Long, Short, and ETF Portfolios



This chart compares the monthly performance of the long portfolio, short portfolio, and the ETF. The long portfolio generally achieved higher returns than the short portfolio, indicating that stocks with stronger momentum performed better.

Chart 3: Cumulative Returns Over Time



This chart shows how returns accumulated over time for both the momentum strategy and the ETF. It highlights differences in long-term growth and shows how volatility affected performance.

The momentum strategy experienced both profitable and unprofitable months. Although the strategy generated positive cumulative returns, the ETF achieved stronger overall performance. The momentum approach also showed higher fluctuations in returns.

Limitations- This study does not account for transaction costs, taxes, or limits on short selling. Because of this, real-world performance may differ from the backtest results.

Final Conclusion This analysis shows that momentum strategies can help identify strong and weak stocks, but they do not always outperform a simple ETF investment. Momentum strategies may offer opportunities, but they also come with greater risk. Which is why people do not use this for their own investments.